Department of Civil, Environmental and Geomatic Engineering

Personality-Driven Crowd Simulation in Indoor Environments with Deep Reinforcement Learning

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1 Introduction

UCL

- *Background*

Crowd Simulation: With the acceleration of urbanization, crowd simulation has become an essential computational tool. It offers a powerful way to analyze and forecast human behavior, particularly in the design and management of **indoor environments** such as shopping malls and galleries.



- *Bridging the Gap in Crowd Simulation*

Current Deep Reinforcement Learning (DRL) Crowd Simulation Models

Limitations:
- Agents are purely physical, neglecting social/psychological factors
- Overlook individual differences, leading to homogeneous behaviors

Personality-Driven Crowd Simulation in Indoor Environments with Deep Reinforcement Learning (P-CSim)

Key Innovation:
- Personalized Agent Definition
- Socialized State Space
- Personality-Behavior Mapping in reward function

- *Research questions*

- RQ1: To what extent do personality parameters influence overall crowd simulation performance metrics such as goal achievement rate, path efficiency, and convergence speed?
- RQ2: To what extent do agents with different personality profiles exhibit divergent behaviors and performance outcomes in crowd simulation?

2 Literature review

2.1 Traditional & Data-Driven Approaches

• **Force-based:** Social Force Model (Helbing & Molnár, 1995)
• **Data-Driven:** Social-LSTM (Alahi et al., 2016) & Social-GAN (Gupta et al., 2018)

Key Strengths:
• Physical realism like collision avoidance
• Path planning

Key Limitations:
• Lack of psychological realism
• Lack dynamic learning

2.2 Social-Psychological Approaches

• **Personality triats**: Big Five Personality Traits (Durupinar et al., 2011)
• **Emotional models:** OCC & PAD (Durupinar et al., 2016)

Key Strengths:
• Enhanced realism
• Social dynamics

Key Limitations:
• Relies on subjective rule-based mapping
• Lack dynamic learning

2.3 Reinforcement Learning-Based Approaches

• **Reinforcement Learning (RL)** (Casadiego and Pelechano, 2015)
• **Deep Reinforcement Learning (DRL)** (Haworth et al., 2020)

Key Strengths:
• Dynamic learning
• Goal-oriented optimization

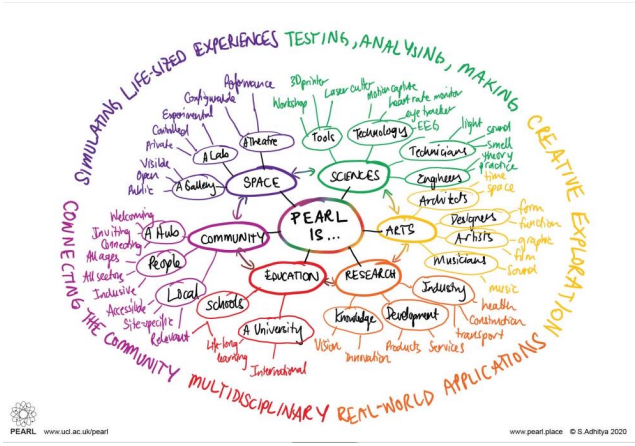
Key Limitations:
• Overlooks social & psychological factors
• Lack behavioral diversity

Personality-Driven Crowd Simulation in Indoor Environments with Deep Reinforcement Learning (P-CSim)

3.1 Data Source

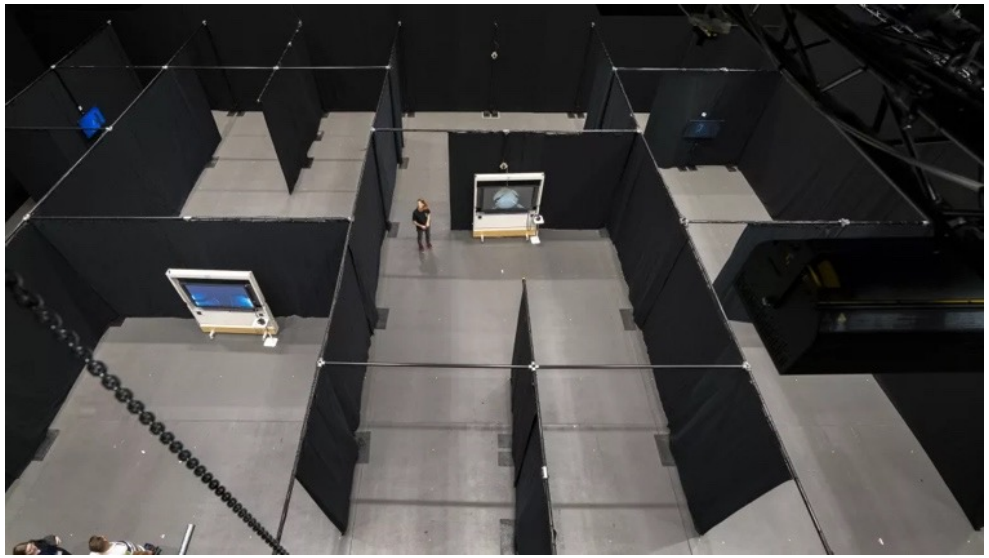
The Person–Environment–Activity Research Laboratory (PEARL)





- A unique, large-scale research facility at University College London (UCL)
- Studies how people interact with their environment and with each other

The 100 Minds in Motion Studies

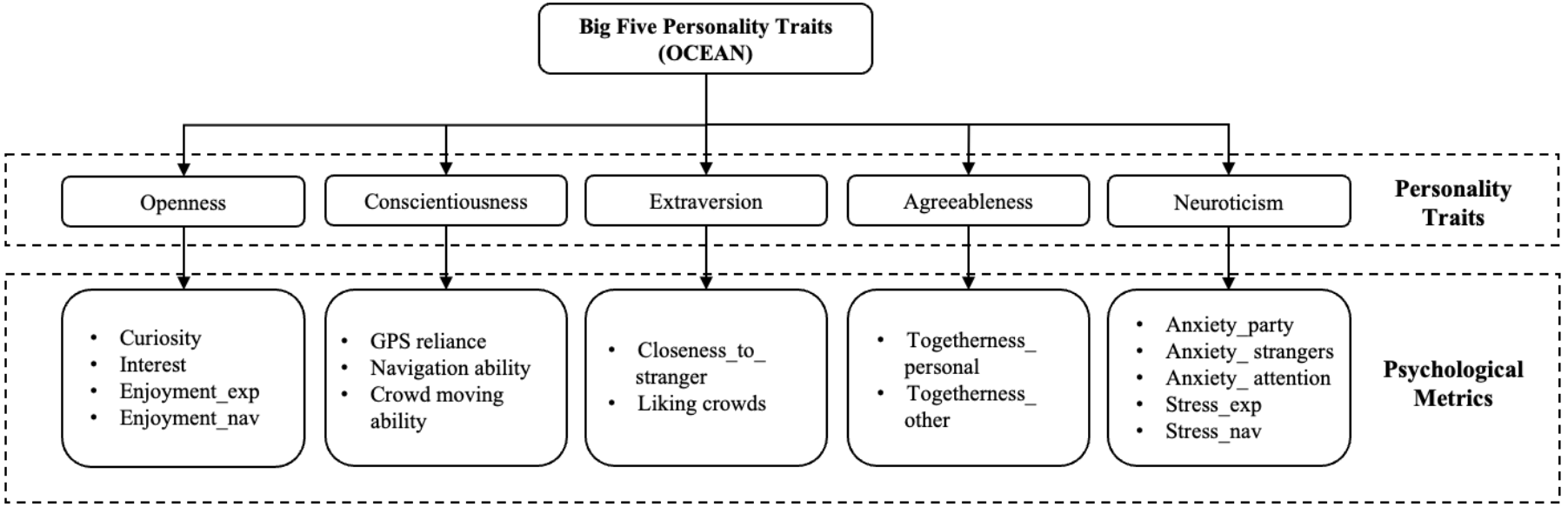


- Investigates **how the built environment** and **individual differences** (e.g., age, mobility, neurodiversity) shape crowd behavior
- Employs **a highly controlled, reconfigurable environment** with advanced motion-tracking cameras and biosensors

This study used **questionnaire data (e.g., age, GPS reliance, curiosity)** and **trajectory data** from 110 participants in the *100 Minds in Motion* studies.

3.2 Data Processing

To establish the relationship between personality traits and behavioral patterns as a foundation for our personality–behavior mapping design



Mapping between Big Five Personality Traits and Psychological Metrics

Process:

1. Calculated each participant's **Big Five personality scores** from questionnaire data

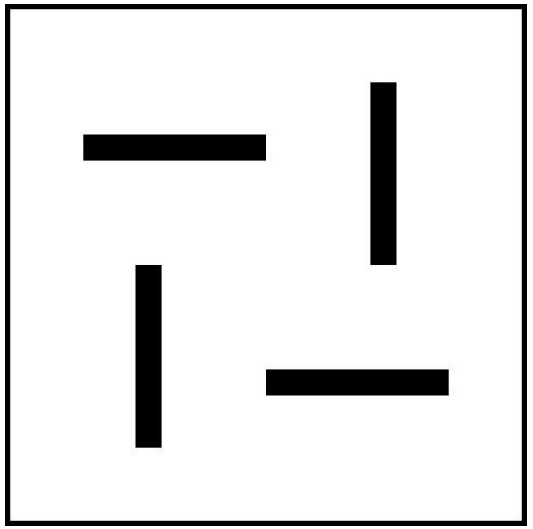
2. Extracted multiple **behavioral metrics** from trajectory data, such as trajectory range and standard deviation of speed

3. Conducted a **correlation analysis** between personality scores and behavioral metrics to identify key relationships

Key Findings:

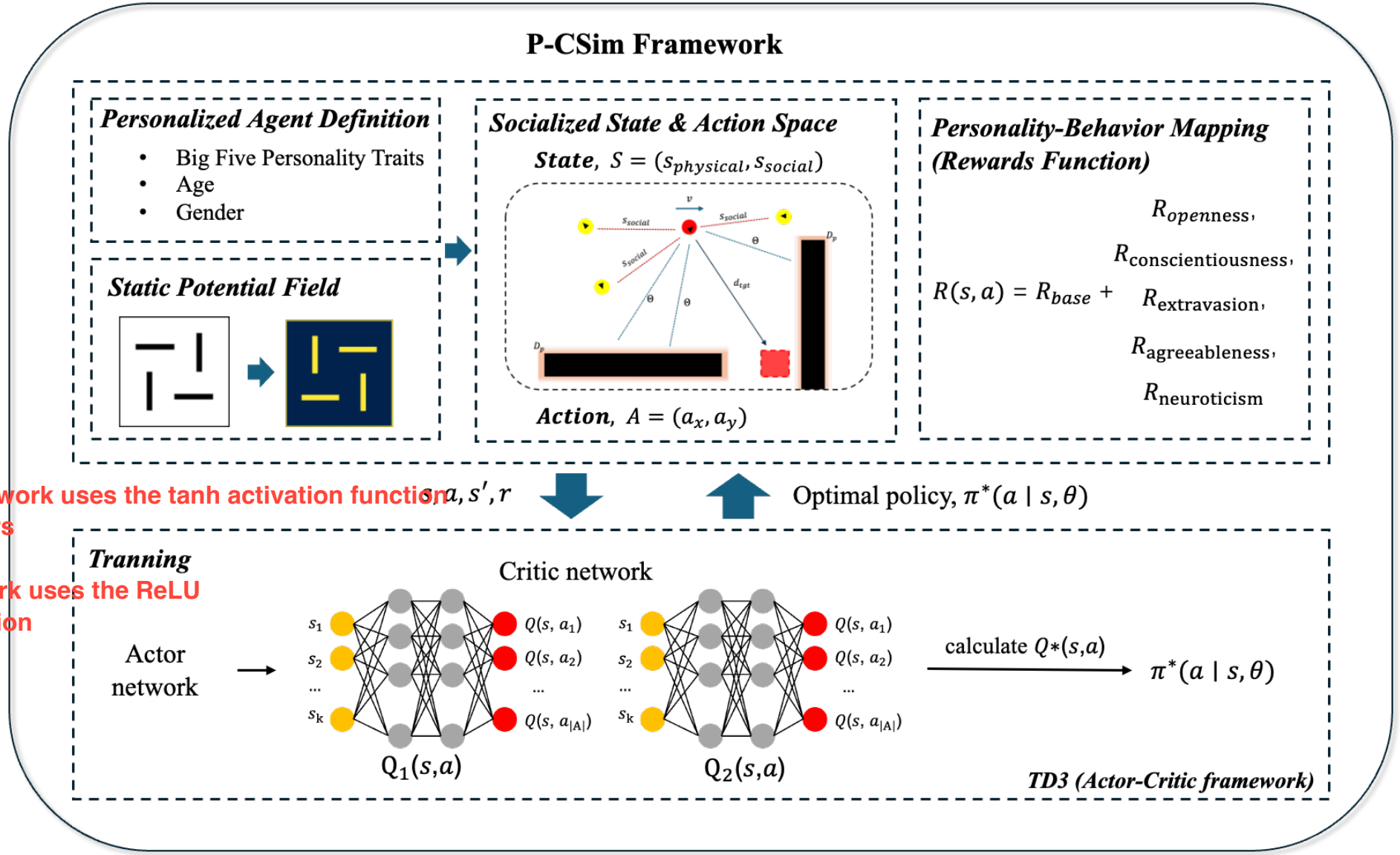
- Openness ↔ Location entropy (r = 0.247)
- Conscientiousness ↔ Max speed (r = 0.159)
- Extraversion ↔ Trajectory range (r = 0.209)
- Neuroticism ↔ Speed std (r = 0.164)

3.3 Simulation Environment



- A simplified 100x100 binary (0-1) grid map
- 0 - traversable space
- 1 - obstacle

4.1 Overall Research Framework (Improved from (Li et al., 2025))



The Actor network uses the tanh activation function between layers

the Critic network uses the ReLU activation function

Li, Y., Chen, Y., Liu, J., Huang, T., 2025. Efficient crowd simulation in complex environment using deep reinforcement learning. Sci. Rep. 15, 5403. https://doi.org/10.1038/s41598-025-88897-2

4.2 Personalized Agent Definition

Agent Definition: Unlike traditional models, our agents are defined by personality traits, age, and gender

Big Five Personality Traits (OCEAN) dimensions

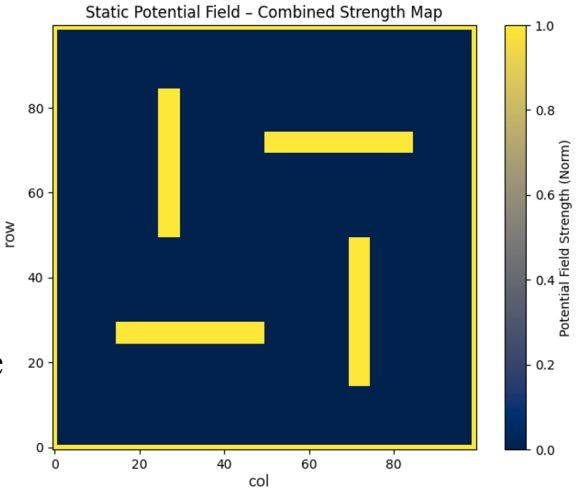
- High Openness / High Conscientiousness / High Extraversion / High Agreeableness / High Neuroticism
- Only one "High" trait (≥0.7), while others are low (e.g., 0.2-0.4)



4.3 Static Potential Field

Method: A static potential field is generated using **Euclidean Distance Transform (EDT)**

- To provide agents with intuitive obstacle information
- The field's values indicate distance to obstacles (smaller values = closer, larger values = farther)



4.4 Socialized State & Action Space

- ***State***, $S = (s_{physical}, s_{social})$

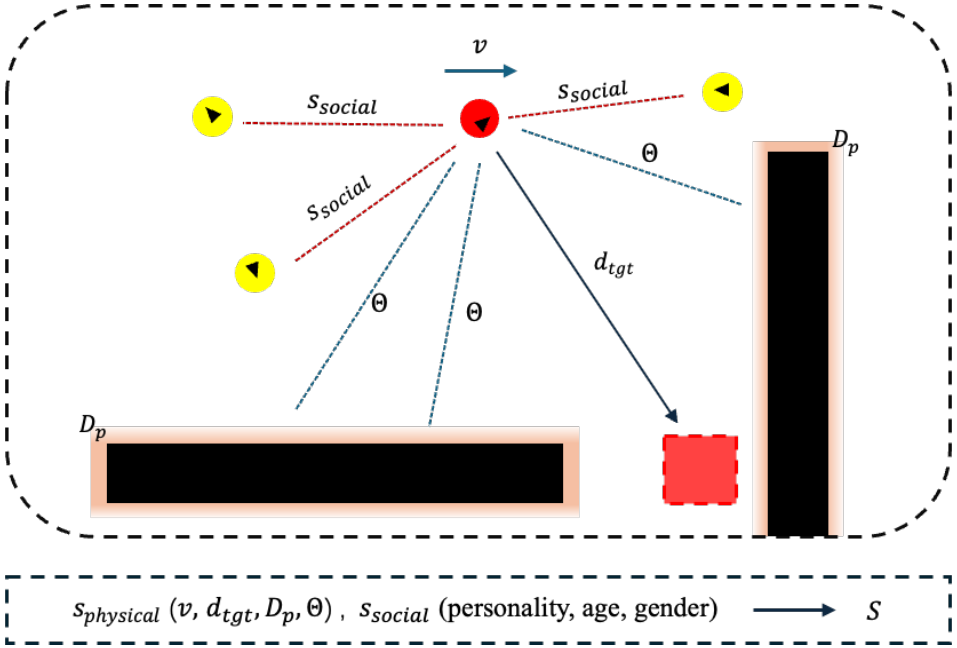
$s_{physical}$ $(v, d_{tgt}, D_p, \Theta)$ ⟶

- v: Agent's velocity
- d_{tgt}: Agent's position
- D_p: The static potential field
- Θ: Agent's obstacle information

s_{social} *(personality, age, gender)*

- ***Action***, $A = (a_x, a_y)$

A continuous 2D acceleration vector



4.5 Personality-Behavior Mapping (Rewards Function) 文本

Through physical rules and reward functions, we connect each agent's personality traits with their behavior learning process. This drives agent to naturally acquire behaviors consistent with their personalities

4.5.1 Openness: We incentivize exploration by rewarding an **increase in the location entropy of the agent's trajectory**

The reward is dynamically calculated based on its *Openness* factor (O) **:**

$$R_{open} = O \times coefficient \times (current_location_entropy - last_location_entropy) \quad (1)$$

$$Location\ entropy: = -\sum_{i=1}^{N} p_i \log_2(p_i) \quad (2)$$

4.5.2 Conscientiousness: We simulate efficiency and task-orientation by assigning a higher **max speed** and weighting **task completion rewards**

The agent's current maximum speed is dynamically calculated based on its *Conscientiousness* factor (C):

$$dynamic_prop_speed = min_speed + (C \times conscientiousness_speed_bonus) \quad (4)$$

4.5.3 Extraversion: We encourage active, social behavior by **expanding the action space** and reducing the **social distance penalty** when near other agents

The social distance penalty is weighted by its *Extraversion* factor (E):

$$P_{social} = P_{social_base} \times (1 - E) \quad (3)$$

4.5.4 Agreeableness: We promote cooperation by enhancing **team objective rewards** and increasing the **penalty for negative social behaviors**

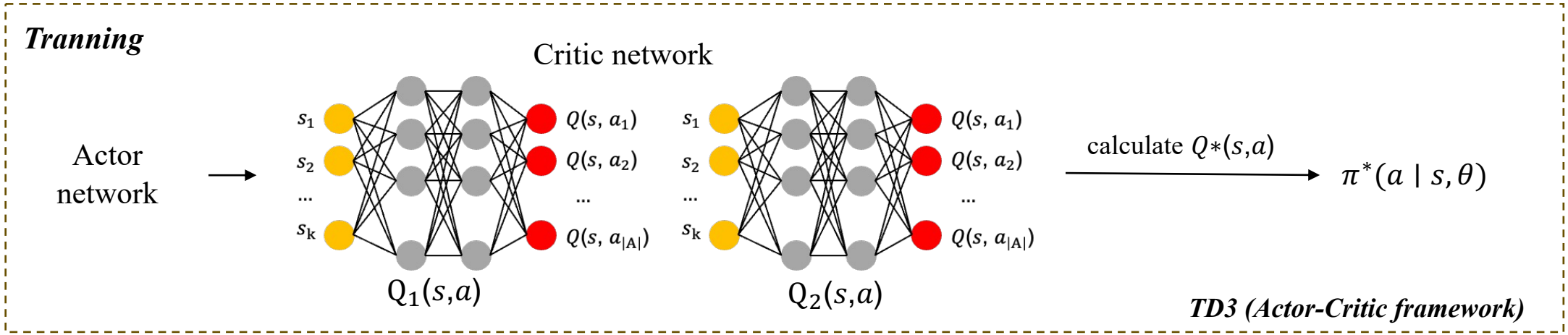
The reward for completing team objectives is weighted by the agent's *Agreeableness* factor (A) as follows:

$$R_{group} = R_{group_base} \times (1 + A \times coefficient) \quad (5)$$

4.5.5 Neuroticism: We model unstable behavior and high sensitivity by introducing **speed standard deviation** in physical rules and **amplifying collision penalties**, the formula for calculating the speed standard deviation is weighted by the agent's *Neuroticism* factor (N) as follows:

$$\sigma = \sigma_{min} + (N \times \sigma_{range}) \quad (6)$$

4.6 Training



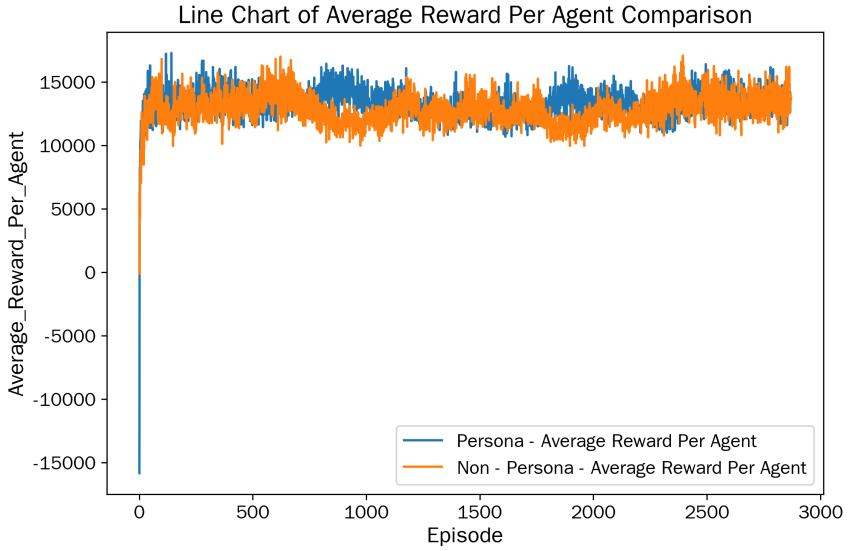
- **Setup:**
  - **20 agents** grouped into **4 teams**
  - Each team consists of **5 agents** with different personality types (OCEAN)

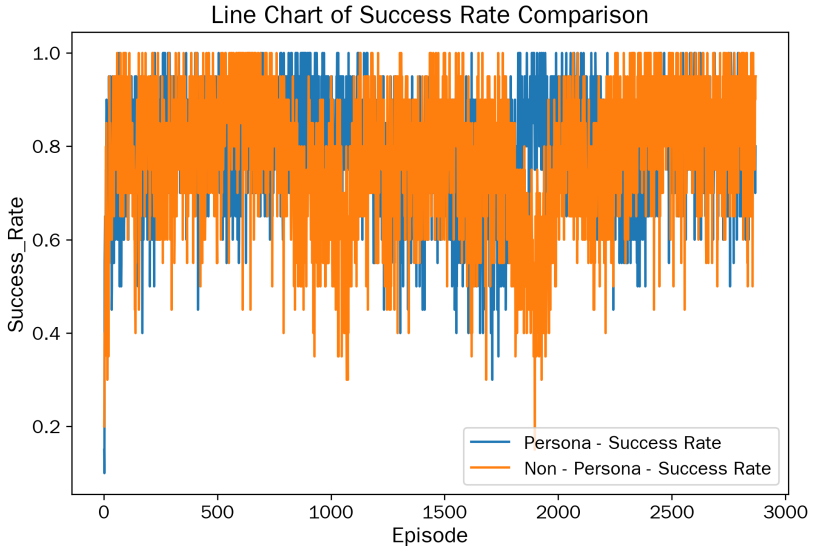
- **Network:**
  - **Actor-Critic framework** with two 1024-neuron hidden layers
  - **The hyperbolic tangent (tanh)** function for the Actor network and **ReLU** for the Critic network
  - **Adam** optimizer is used for network parameter updates

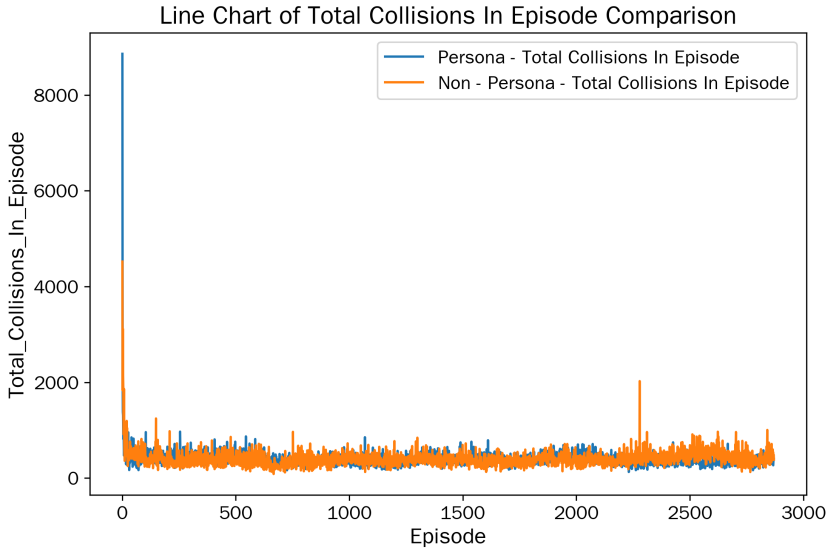
Parameter	Value	Description
γ	0.99	Discount factor
$A_r l$	1e-4	Actor learning rate
$C_r l$	1e-4	Critic learning rate
B_{size}	512	Batch size
RB_{size}	1e6	Reply buffer size
τ	0.005	Target update rate
P_n	0.2	Policy noise
E_n	0.25	Exploration noise

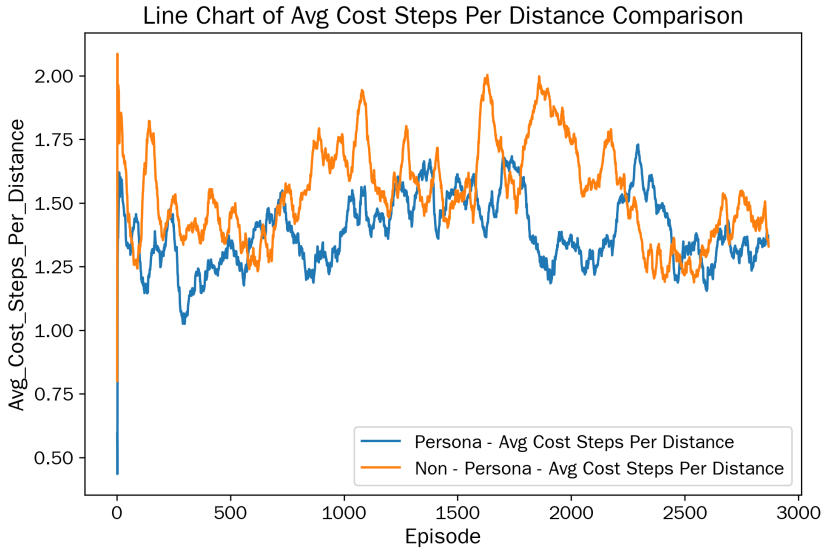
Li, Y., Chen, Y., Liu, J., Huang, T., 2025. Efficient crowd simulation in complex environment using deep reinforcement learning. Sci. Rep. 15, 5403. https://doi.org/10.1038/s41598-025-88897-2

5.1 The Impact of Personality Module Integration on Code Complexity









Key Findings:

- **Training Stability:** Personality agents show **more stable convergence**
- **Interaction Logic:** Statistical tests show **no significant difference** in collision risk ($p > 0.05$)
- **Overall Performance:** The module enhances performance **without significantly adding code complexity or instability**

5.2 Impact of Personality on Crowd Simulation

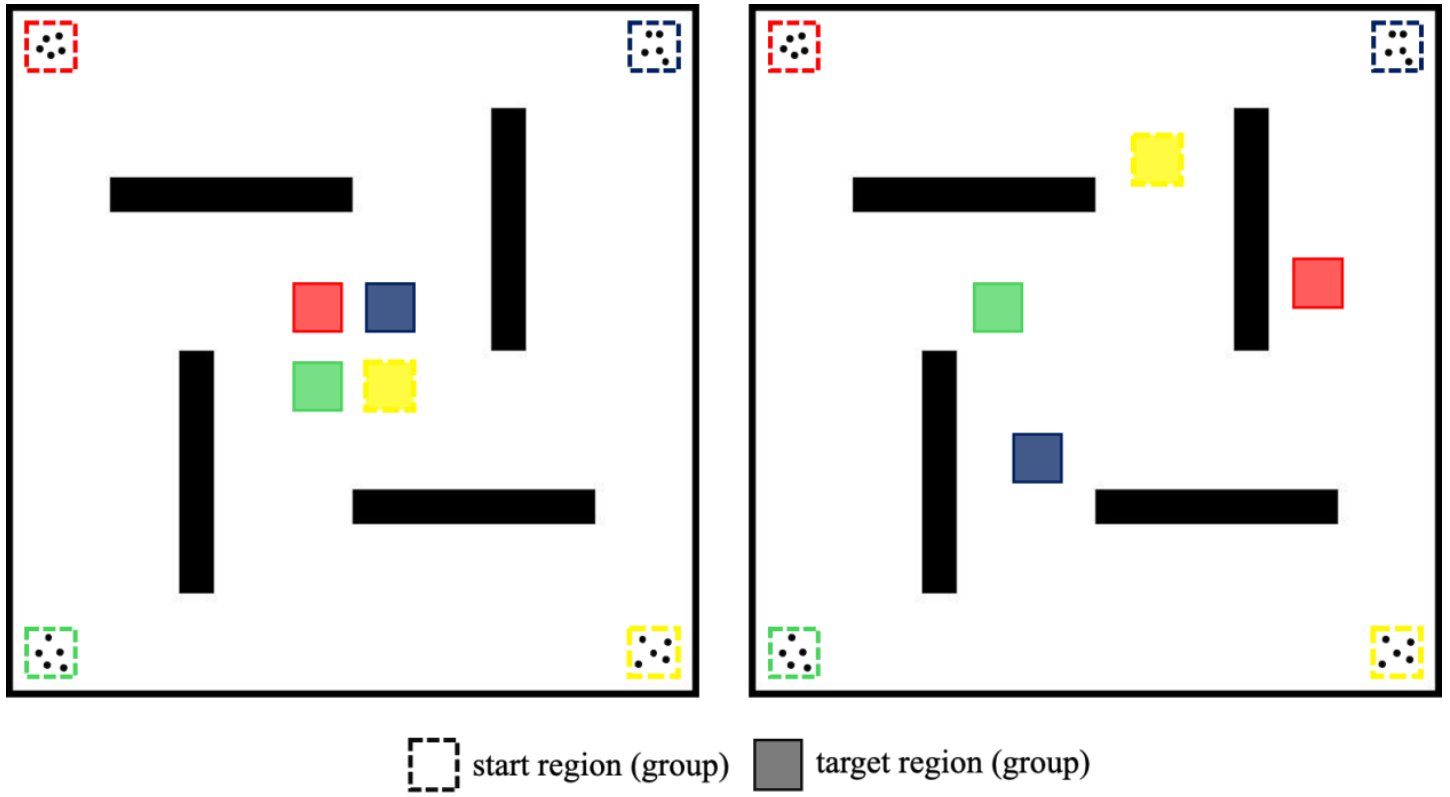
- Evaluation setting

1. Environments

- **Fixed-Target evaluation**: Target is fixed at the center of the map, to evaluate performance in a predictable target
- **Random-Target evaluation** : Target is randomly generated for each episode, to test ability to generalize to unpredictable target

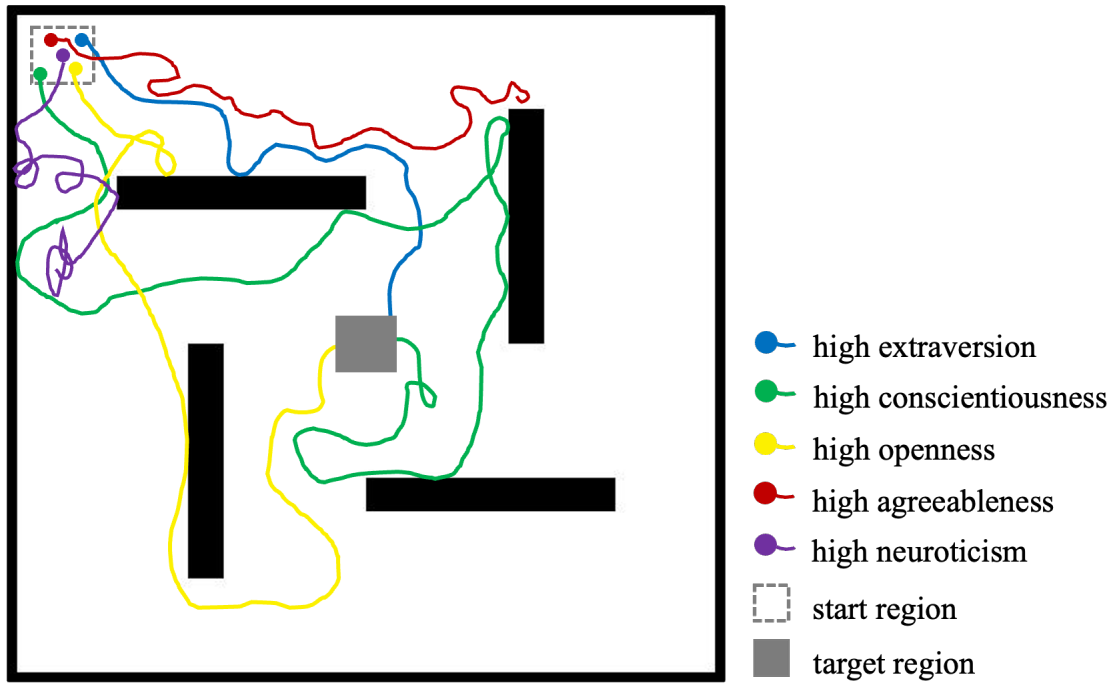
2. Agent Setup: Each mode used **20 agents** (4 teams of 5 personalities each)

3. Data Robustness: The experiments were conducted over **10 independent trials**, and all performance indicators were **averaged across the trials** to ensure reliability

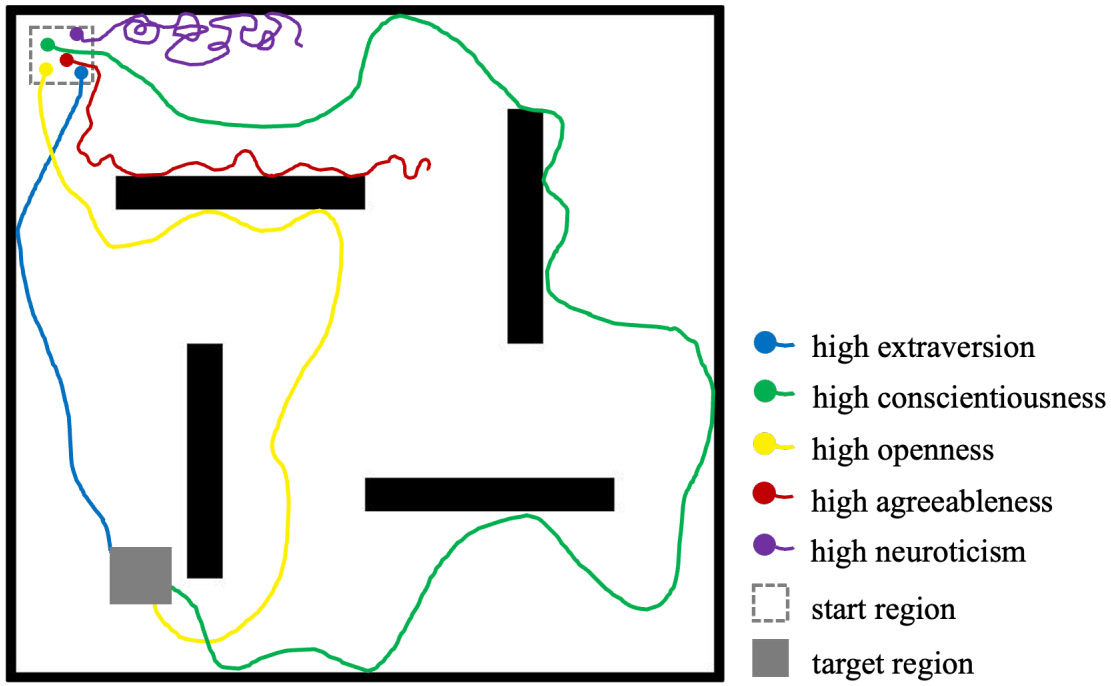


(a) Fixed-target evaluation **(b) Random-target evaluation**

5.2 Impact of Personality on Crowd Simulation



5.2.1 Fixed-target evaluation



5.2.2 Random-target evaluation

High Extraversion:
- **Best Performance**: Highest success rates (**70% & 80%**)
- **Efficient Navigation**: Lowest collisions (**9.1 & 6.95**) and shortest paths (**2.40m & 2.43m**)

High Conscientiousness:
- **Success Rates**: Reached **62.5% & 55%**.
- **Task-Oriented**: Resulted in longer paths and more collisions to ensure task completion

High Openness:
- **Exploratory Behavior**: Showed longer paths and more collisions
- **Success Rates**: Reached **65%** in the fixed-target environment

High Agreeableness & High Neuroticism:
- **Worst Performance**: Lowest success rates (**5% & 22.5%**)
- **Disordered Behavior**: Neuroticism-driven agents had the highest collisions (**136.1 & 139.1**) and most unstable navigation

6 Discussion & Conclusion

6.1 Answering the Research Questions and Key Findings

- **Answer to RQ1:** The integration of the personality module is **feasible** from an engineering perspective. It provides performance enhancements **without a significant increase in core code complexity**, serving as a **lightweight and non-intrusive** extension

- **Answer to RQ2:** Personality traits have a **significant and consistent impact** on agent behavior and task success. We've confirmed a direct mapping between personality and behavior, which makes agent behavior **more realistic and interpretable**
  - **High-success groups:** Agents with high **Extraversion**, **Conscientiousness**, and **Openness** performed better
  - **Low-success groups:** Agents with high **Agreeableness** and **Neuroticism** struggled

6.2 Limitation & Future Work

- **Limitations:**
  - **Simple Environments and Mapping:** The experimental environment and the personality-behavior mapping model were relatively simple
  - **Focus on Individuals:** The study primarily examined the impact of individual personality traits, while giving limited attention to group-level interactions between agents
- **Future Work:**
  - Expand to **more complex environments and crowds**
  - Explore **more advanced, nonlinear behavior mapping** methods
  - Investigate the **dynamic interactions** between agents of different personality types in a group context

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